

Unit 4 Quest

Waves and LightSHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

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Perfect!
Excellent work!

THINKING

1. The highest recorded ocean waves were observed by a U.S. Navy ship in the north Pacific Ocean in 1933. The amplitude was 34 m. The period was 14.8 s, and the wavelength was 342 m. How fast were the waves travelling?

[2 marks]

$$T = 14.8 \text{ s}$$

$$\lambda = 342 \text{ m}$$

$$v = \frac{\lambda}{T}$$

$$v = \frac{342 \text{ m}}{14.8 \text{ s}}$$

$$v = 23 \text{ m/s [F]}$$

Final answer: $v = 23 \text{ m/s [F]}$

2. The depth of the water at a point on a lake can be determined by sending a high-frequency sound wave vertically to the bottom of the lake, and timing how long it takes for the sound waves to return. This is called SONAR, for sound navigation and ranging. A typical SONAR set operates at 192 kHz, and has a wavelength of 0.772 cm. If a signal return 0.264 s after it is sent, how deep is the water?

[3 marks]

$$\lambda = 0.772 \text{ cm} = 0.00772 \text{ m}$$

$$2t = 0.264 \text{ s}$$

$$f = 192 \text{ kHz} = 192000 \text{ Hz}$$

$$t = 0.132 \text{ s}$$

$$v = f\lambda$$

$$v = 192000 \text{ Hz} \cdot 0.00772 \text{ m}$$

$$v = 1482.24 \text{ m/s [D]}$$

$$d = \frac{v}{t}$$

$$d = vt$$

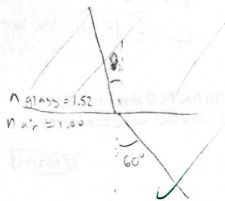
$$d = 1482.24 \text{ m/s} \cdot 0.132 \text{ s}$$

$$\therefore d = 195.66 \text{ m [D]}$$

Final answer: $d = 195.7 \text{ m [D]}$

3. Light travels from crown glass ($n_{\text{glass}} = 1.52$) into air ($n_{\text{air}} = 1.00$). The angle of refraction in air is 60.0° . Calculate the angle of incidence in the crown glass.

[2 marks]



$$n_g = 1.52 = n_1$$

$$n_a = 1.00 = n_2$$

$$\theta_2 = 60^\circ$$

$$\theta_1 = ?$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\sin \theta_1 = \frac{n_2 \sin \theta_2}{n_1}$$

$$\theta_1 = \sin^{-1} \left(\frac{n_2 \sin \theta_2}{n_1} \right)$$

$$\theta_1 = \sin^{-1} \left(\frac{1.0 \sin 60^\circ}{1.52} \right)$$

$$\theta_1 = 34.73^\circ$$

$$\therefore \theta_1 = 34.7^\circ$$

Final answer: $\theta_1 = 34.7^\circ$

4. A straight boundary separates two bodies of rock. Longitudinal earthquake waves, travelling through the first body at 7.75 km/s , meet the boundary at an angle of incidence of 20.0° . The wave speed in the second body is 7.72 km/s . Calculate the angle of refraction.

[2 marks]

$$v_1 = 7.75 \text{ km/s} = 7750 \text{ m/s}$$

$$v_2 = 7.72 \text{ km/s} = 7720 \text{ m/s}$$

$$\theta_1 = 20^\circ$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}$$

$$\sin \theta_2 = \frac{v_2 \sin \theta_1}{v_1}$$

$$\theta_2 = \sin^{-1} \left(\frac{v_2 \sin \theta_1}{v_1} \right)$$

$$\theta_2 = \sin^{-1} \left(\frac{7720 \sin 20^\circ}{7750} \right)$$

$$\therefore \theta_2 = 19.92^\circ$$

Final answer: $\theta_2 = 19.9^\circ$

5. Monochromatic light falling on two slits 0.018 mm apart produces the fifth-order bright fringe at an 8.6° angle. What is the wavelength of the light used?

(2 marks)

$$d = 0.018 \text{ mm} = 1.8 \times 10^{-5} \text{ m}$$

$$m = 5$$

$$\theta = 8.6^\circ$$

$$\lambda = ?$$

$$d \sin \theta = m \lambda$$

$$\frac{d \sin \theta}{m} = \lambda$$

$$\lambda = \frac{(1.8 \times 10^{-5}) \cdot (\sin 8.6^\circ)}{5}$$

$$\therefore \lambda = 5.38 \times 10^{-7} \text{ m}$$

Final answer: $\lambda = 5.38 \times 10^{-7} \text{ m}$

6. A parallel beam of light from a Helium-neon laser, with a wavelength 633 nm, falls on two very narrow slits 0.068 mm apart. How far apart are the fringes in the center of the pattern on a screen 3.3 m away?

(3 marks)

$$\lambda = 633 \text{ nm} = 6.33 \times 10^{-7} \text{ m}$$

$$d = 0.068 \text{ mm} = 6.8 \times 10^{-5} \text{ m}$$

$$L = 3.3 \text{ m}$$

$$\lambda = x \left(\frac{d}{L} \right)$$

$$x = \frac{\lambda L}{d}$$

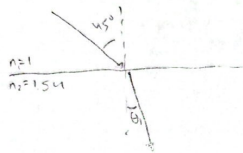
$$x = \frac{6.33 \times 10^{-7} \cdot 3.3 \text{ m}}{6.8 \times 10^{-5}}$$

$$\therefore x = 0.031 \text{ m}$$

Final answer: $x = 0.031 \text{ m}$

7. Light is incident on an equilateral glass prism at a 45.0° angle to one face as shown in the diagram. Calculate the angle at which light emerges from the opposite face. Assume that $n = 1.54$. Show all your work. [4 marks]

face 1:



$n_1 = 1$

$n_2 = 1.54$

$\theta_1 = 45.0^\circ$

$\theta_2 = ?$

$n_1 \sin \theta_1 = n_2 \sin \theta_2$

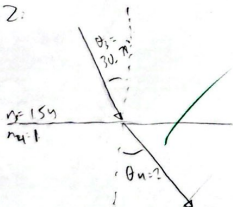
$\sin \theta_2 = \frac{n_1 \sin \theta_1}{n_2}$

$\theta_2 = \sin^{-1} \left(\frac{n_1 \sin \theta_1}{n_2} \right)$

$\theta_2 = \sin^{-1} \left(\frac{1 \cdot \sin 45}{1.54} \right)$

$\theta_2 = 27.33^\circ$

face 2:



$n_3 = 1.54$

$\theta_3 = 32.667^\circ$

$n_4 = 1$

$\theta_4 = ?$

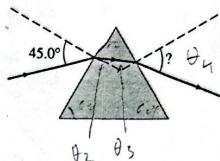
$n_3 \sin \theta_3 = n_4 \sin \theta_4$

$\sin \theta_4 = \frac{n_3 \sin \theta_3}{n_4}$

$\theta_4 = \sin^{-1} \left(\frac{n_3 \sin \theta_3}{n_4} \right)$

$\theta_4 = \sin^{-1} \left(\frac{1.54 \cdot \sin 32.667^\circ}{1} \right)$

$\therefore \theta_4 = 56.2^\circ$



$\theta_3 = 60^\circ - \theta_2$

$60^\circ - 27.33^\circ = 32.67^\circ$

$\theta_3 = 32.667^\circ$

Final answer: $\theta_4 = 56^\circ$